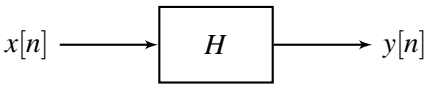
# EE 66 Signals, Dynamics, and Information
# Fall 2026 Note 12

## 12.1 LTI Systems

A discrete-time Linear Time-Invariant (LTI) system maps an input sequence $x[n]$ to an output sequence $y[n]$ and can be described by a **difference equation**. For example, $y[n] - 0.5y[n-1] = x[n]$ updates the present output using the present input and past output.




Two properties define LTI systems:

- **Linearity:** A system is linear if it obeys homogeneity and additivity.
  - **Homogeneity:** if $x[n] \to y[n]$, then $\alpha x[n] \to \alpha y[n]$ for any scalar $\alpha$.
  - **Additivity:** if $x_1[n] \to y_1[n]$ and $x_2[n] \to y_2[n]$, then $x_1[n] + x_2[n] \to y_1[n] + y_2[n]$.

  Together these give **superposition**: for any scalars $\alpha, \beta$,

  $$\alpha x_1[n] + \beta x_2[n] \to \alpha y_1[n] + \beta y_2[n].$$

  Linearity indicates that inputs to the system do not interact with each other, such that the output is fully predictable by their individual responses.

- **Time Invariance:** Shifts commute with the system. If $x[n] \to y[n]$, then $x[n-k] \to y[n-k]$ for any integer shift $k$. Time invariance indicates that the system itself does not change over time.

**Example 1.** Is the system $y[n] = 3x[n] - x[n-1]$ LTI?

*Linearity.* Let $x_1[n] \to y_1[n]$ and $x_2[n] \to y_2[n]$, and consider the input $x_3[n] = \alpha x_1[n] + \beta x_2[n]$. Its output is

$$\begin{aligned} y_3[n] &= 3x_3[n] - x_3[n-1] \\ &= 3\big(\alpha x_1[n] + \beta x_2[n]\big) - \big(\alpha x_1[n-1] + \beta x_2[n-1]\big) \\ &= \alpha\big(3x_1[n] - x_1[n-1]\big) + \beta\big(3x_2[n] - x_2[n-1]\big) \\ &= \alpha y_1[n] + \beta y_2[n]. \end{aligned}$$

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The system is linear.

*Time invariance.* Let $x[n] \to y[n]$, and consider the delayed input $x_d[n] = x[n-k]$ with output $y_d[n]$. Then

$$y_d[n] = 3x_d[n] - x_d[n-1] = 3x[n-k] - x[n-k-1],$$
$$y[n-k] = 3x[n-k] - x[(n-k)-1] = 3x[n-k] - x[n-k-1].$$

Since $y_d[n] = y[n-k]$, the system is time-invariant. Therefore the system is LTI.

**Example 2.** Is the system $y[n] = x[n] + n$ LTI?

*Linearity.* With $x_1 \to y_1$, $x_2 \to y_2$, and $x_3[n] = \alpha x_1[n] + \beta x_2[n]$, we compare

$$y_3[n] = x_3[n] + n = \alpha x_1[n] + \beta x_2[n] + n,$$
$$\alpha y_1[n] + \beta y_2[n] = \alpha x_1[n] + \beta x_2[n] + (\alpha + \beta)\, n.$$

These are not in general equal, so the system is *not* linear.

*Time invariance.* We've already shown that the system is not LTI, but let's check time-invariance for practice. With $x_d[n] = x[n-k]$ and output $y_d[n]$,

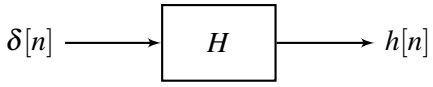
$$y_d[n] = x_d[n] + n = x[n-k] + n,$$
$$y[n-k] = x[n-k] + (n-k).$$

Since $y_d[n] \neq y[n-k]$, the system is *not* time-invariant either.

The zero input offers a faster way to see that this system is not linear. With $x[n] = 0$ the output is $y[n] = n$, not the zero signal. A linear system can never do this: taking $\alpha = 0$ in homogeneity gives $0 \cdot x[n] \to 0 \cdot y[n]$, so the zero input must produce the zero output.

# 12.2 Impulse Response

In this section we will demonstrate that LTI systems can be *fully* characterized by their response to an impulse signal. An impulse is defined as the Kronecker delta $\delta[n]$, which has a value of one at $n = 0$ and zero elsewhere.




$h[n]$ is defined as the system's output to a $\delta[n]$ input, also known as the **impulse response** of the system.

An important observation is that any sequence (for example, discrete-time signals) can be written as a weighted sum of shifted deltas:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]. \tag{1}$$

Each term $x[k]\,\delta[n-k]$ is a spike of height $x[k]$ at time $k$, so at any time $n$ only the term with $k=n$ survives and the sum equals $x[n]$.

Now we pass each piece of this decomposition through the system.

$$\begin{aligned}
\delta[n] &\;\to\; h[n] && \text{(definition of } h[n]\text{)}\\
\delta[n-k] &\;\to\; h[n-k] && \text{(time invariance)}\\
x[k]\,\delta[n-k] &\;\to\; x[k]\,h[n-k] && \text{(homogeneity; } x[k] \text{ is just a constant)}\\
\sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k] &\;\to\; \sum_{k=-\infty}^{\infty} x[k]\,h[n-k] && \text{(additivity)}
\end{aligned}$$

By (1), the input on the last line is just $x[n]$, so the output is

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]. \tag{2}$$

## 12.2.1 A complete descriptor

By the property that any input can be decomposed into shifted impulses, the impulse response $h[n]$ completely characterizes the LTI system. Knowing $h[n]$ allows us to compute the output for any input $x[n]$ using the formula above. In other words, a single measurement of the system, its response to one impulse, is enough to predict its response to every input.

## 12.2.2 Difference equations

For LTI systems defined by a linear constant-coefficient difference equation such as

$$y[n] - a_1 y[n-1] - \cdots - a_N y[n-N] = b_0 x[n] + \cdots + b_M x[n-M],$$

the impulse response is the solution when $x[n]=\delta[n]$ and $y[n]=0$ for $n<0$. Solving once gives $h[n]$, and every other input follows by superposition and shifting.

**Example.** Find the impulse response of $y[n] - 0.5\,y[n-1] = x[n]$.

Set $x[n]=\delta[n]$ and write the equation as a recursion, $h[n] = 0.5\,h[n-1] + \delta[n]$, with $h[n]=0$ for $n<0$. Then

$$h[0] = 0.5\cdot 0 + 1 = 1, \qquad h[1] = 0.5\cdot 1 = 0.5, \qquad h[2] = 0.5\cdot 0.5 = 0.25, \;\ldots$$

Notice that $h[n] = 0.5^n\,u[n]$, where $u[n]$ is the unit step (one for $n \geq 0$, zero otherwise). The same argument for the general case of $y[n] - a\,y[n-1] = x[n]$ gives $h[n] = a^n u[n]$.

# 12.3 Convolution

Convolution ($*$) describes the input-output relation of LTI systems derived above. The output $y[n]$ is the convolution of the input $x[n]$ with the impulse response $h[n]$:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k] = (x*h)[n]. \tag{3}$$

## 12.3.1 Flip-and-slide

The term $h[n-k]$, viewed as a function of the summation index $k$, is the impulse response **flipped** in time and then **slid** to position $n$. This gives a graphical recipe for computing convolution:

1. **Flip.** Reflect $h[k]$ about the origin to get $h[-k]$.
2. **Slide.** Shift the result right by $n$ to get $h[n-k]$. The sample $h[0]$ now sits at $k=n$.
3. **Multiply and sum.** Multiply $x[k]$ by $h[n-k]$ at every index $k$ and sum. This gives $y[n]$.
4. **Repeat** for each $n$. As $n$ increases, $h[n-k]$ sweeps across $x[k]$ from left to right.

The flipped impulse response acts as a sliding set of **weights**: at each position it computes a weighted sum of the input, much like a moving weighted average. The output is nonzero only where the $x[k]$ and $h[n-k]$ overlap; for finite sequences of lengths $L_x$ and $L_h$ the output has length $L_x+L_h-1$.

**Example.** Let $x[n]=1,2,3$ at $n=0,1,2$ and $h[n]=2,1$ at $n=0,1$ (zero elsewhere). Flipping $h$ gives the weights $1,2$ (with $h[0]=2$ on the right). The table aligns $x[k]$ with $h[n-k]$ for each output time $n$:

| | $k$=−1 | $k$=0 | $k$=1 | $k$=2 | $k$=3 | |
|---|---|---|---|---|---|---|
| $x[k]$ | 0 | 1 | 2 | 3 | 0 | |
| $h[0-k]$ | 1 | 2 | 0 | 0 | 0 | $y[0]=1\cdot 2=\mathbf{2}$ |
| $h[1-k]$ | 0 | 1 | 2 | 0 | 0 | $y[1]=1\cdot 1+2\cdot 2=\mathbf{5}$ |
| $h[2-k]$ | 0 | 0 | 1 | 2 | 0 | $y[2]=2\cdot 1+3\cdot 2=\mathbf{8}$ |
| $h[3-k]$ | 0 | 0 | 0 | 1 | 2 | $y[3]=3\cdot 1=\mathbf{3}$ |

The output $y[n]=2,5,8,3$ for $n=0,\ldots,3$ has length $3+2-1=4$, as expected. Figure 1 shows the same computation graphically.

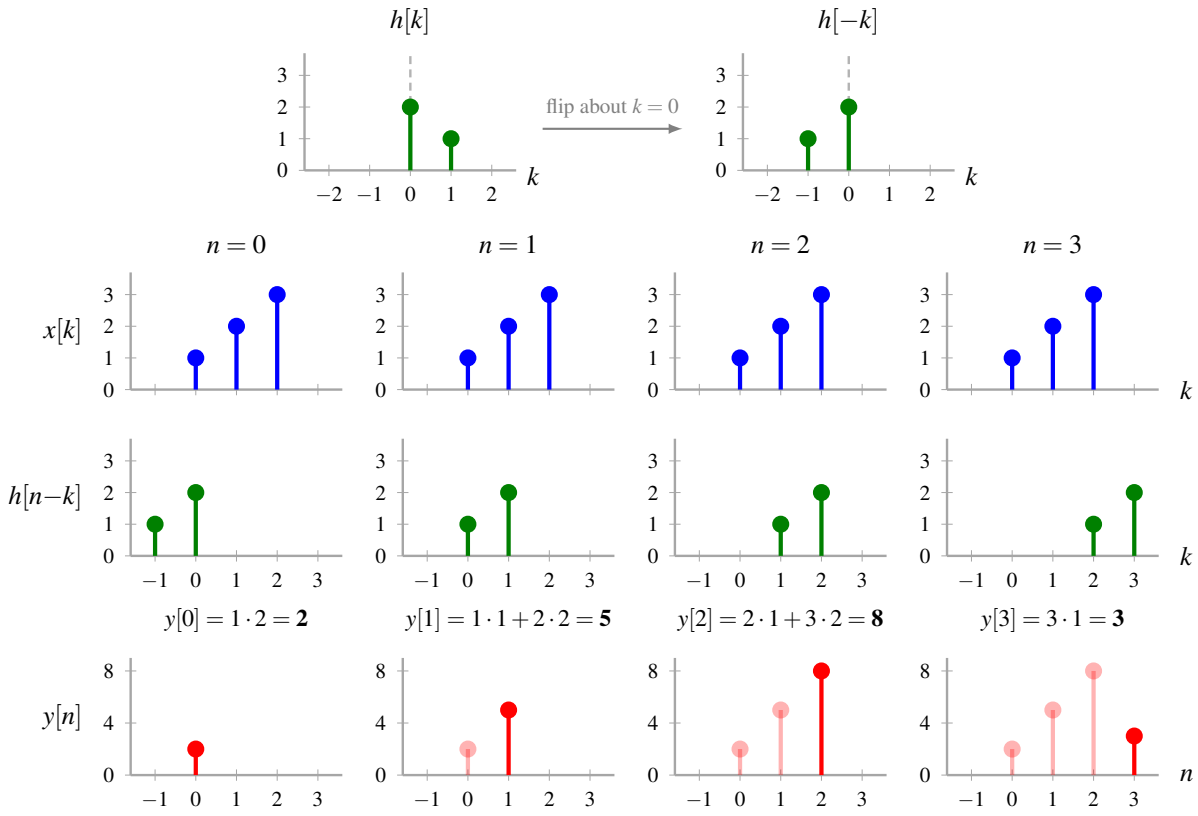



**Figure 1:** Flip-and-slide for $x = (1,2,3)$, $h = (2,1)$. **Top:** $h[k]$ is flipped to give $h[-k]$. **Columns:** for each $n$, $h[n-k]$ (green) slides right while $x[k]$ (blue) stays fixed; overlapping samples are multiplied and summed. **Bottom:** the output $y[n]$ (red) builds up one sample per shift.

## 12.3.2 Convolution properties

Important properties follow directly from the mathematical definition of convolution:

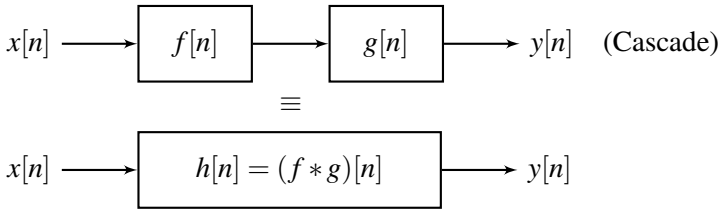
- **Commutativity:** Swapping $x$ and $h$ does not change the result. Let $m = n - k$:

$$(x * h)[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k] = \sum_{m=-\infty}^{\infty} x[n-m]\,h[m] = (h * x)[n].$$

- **Associativity:** Grouping does not matter. Let overall impulse response $h = f * g$. Expanding $((x*f)*g)[n]$:

$$\begin{aligned}((x*f)*g)[n] &= \sum_{m=-\infty}^{\infty}\left(\sum_{k=-\infty}^{\infty} x[k]\,f[m-k]\right) g[n-m] \\ &= \sum_{k=-\infty}^{\infty} x[k] \underbrace{\sum_{m=-\infty}^{\infty} f[m-k]\,g[n-m]}_{(f*g)[n-k]=h[n-k]} \\ &= \sum_{k=-\infty}^{\infty} x[k]\,h[n-k] \\ &= (x*h)[n].\end{aligned}$$

  Cascaded LTI systems defined by $f$ and $g$ can be replaced by a single system $h$.




- **Distributivity:** Convolution distributes over addition:

$$(x*(f+g))[n] = \sum_{k=-\infty}^{\infty} x[k]\,(f+g)[n-k] = (x*f)[n] + (x*g)[n].$$

- **Identity:** The delta leaves sequences unchanged:

$$(x*\delta)[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k] = x[n].$$

- **Identity:** The delta leaves sequences unchanged:

$$(x*\delta)[n] = \sum_{k} x[k]\,\delta[n-k] = x[n].$$

- **Shifts:** A shifted delta shifts the output:

$$(x*\delta_N)[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta_N[n-k] = x[n-N], \quad \text{where } \delta_N[n] \triangleq \delta[n-N].$$

### 12.3.3 Practical examples

To further build intuition, let us examine three canonical examples of convolution. These examples highlight how the choice of impulse response $h[n]$ dictates the behavior of the system.

### 12.3.3.1 Example 1: convolution of rectangular pulses

Consider two rectangular pulses of length $L = 2$. Let the input be $x[n] = \delta[n] + \delta[n-1]$ and the impulse response be $h[n] = \delta[n] + \delta[n-1]$.

When we convolve these finite sequences, the non-zero duration of the output is determined by $L_y = L_x + L_h - 1 = 2 + 2 - 1 = 3$.

$$\begin{aligned} y[n] &= (x * h)[n] \\ &= x[n] * (\delta[n] + \delta[n-1]) \\ &= x[n] + x[n-1] \\ &= \big(\delta[n] + \delta[n-1]\big) + \big(\delta[n-1] + \delta[n-2]\big) \\ &= \delta[n] + 2\delta[n-1] + \delta[n-2]. \end{aligned}$$

Notice that convolving two rectangular pulses results in a triangular pulse.

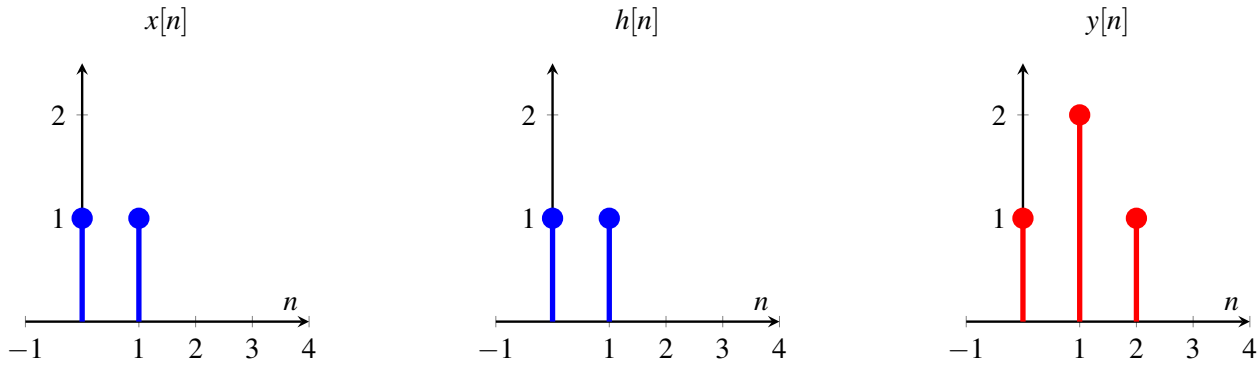



**Figure 2:** Visualizing the convolution of two rectangular pulses. Note that the output duration ($L = 3$) equals the sum of input durations minus one.

### 12.3.3.2 Example 2: convolution with a unit step (the accumulator)

Consider an arbitrary finite signal $x[n]$ convolved with the unit step $h[n] = u[n]$.

$$y[n] = (x * h)[n] = (x * u)[n] = \sum_{k=-\infty}^{\infty} x[k]u[n-k]$$

Since $u[n-k] = 1$ only when $n - k \geq 0$ (equivalent to $k \leq n$) and is 0 otherwise, the upper limit of the summation becomes $n$:

$$y[n] = \sum_{k=-\infty}^{n} x[k]$$

Convolving a signal with a unit step creates a **running sum** (or accumulation) of that signal. This is the discrete-time equivalent of integration.

For example, let $x[n] = \delta[n] - \delta[n-2]$. This signal has a value of $+1$ at $n = 0$ and $-1$ at $n = 2$. The output accumulates these values: it steps up to 1 at $n = 0$, stays there at $n = 1$, and steps back down to 0 at $n = 2$.

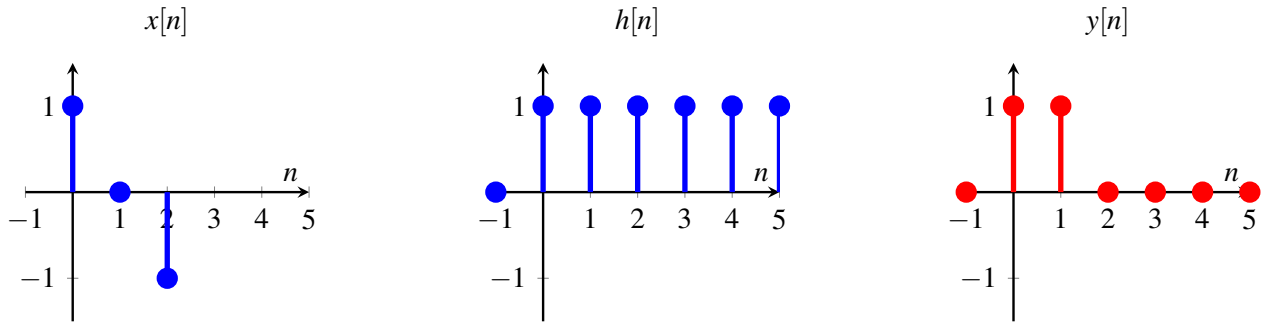



**Figure 3:** Visualizing the Accumulator property. Convolving with $u[n]$ integrates the input signal.

### 12.3.3.3 Example 3: unit step with unit step (the ramp)

What happens if we put a unit step $x[n] = u[n]$ into the accumulator system with impulse response $h[n] = u[n]$?

Using the accumulator property derived in Example 2:

$$y[n] = \sum_{k=-\infty}^{n} u[k]$$

For $n < 0$, the sum is zero. For $n \geq 0$, we sum the value 1 from $k = 0$ to $n$, which implies we are summing $(n+1)$ ones.

$$y[n] = \begin{cases} 0 & n < 0 \\ n+1 & n \geq 0 \end{cases} = (n+1)u[n]$$

This function is known as the discrete-time **ramp**.

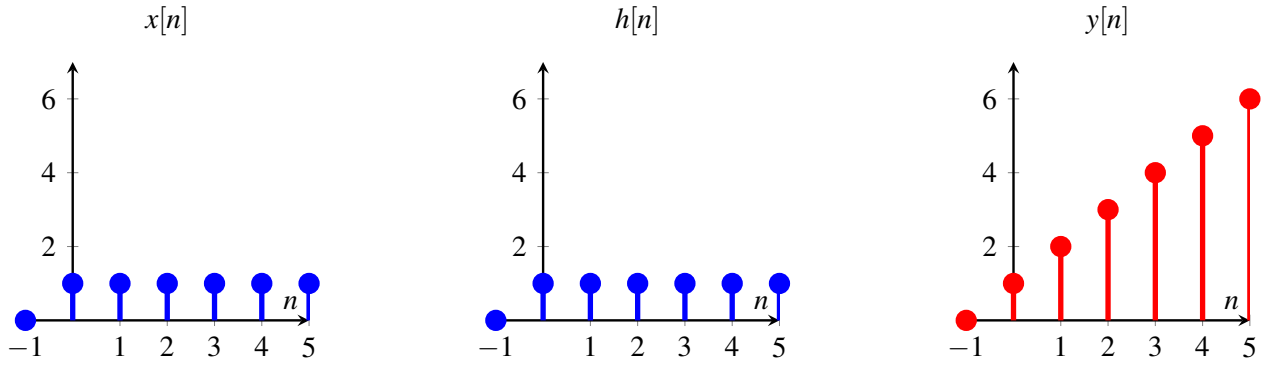



**Figure 4:** Visualizing the convolution of two unit steps. The accumulation of constant 1s results in a linear ramp.

### 12.3.4 Stability

In Example 3, the input $u[n]$ never exceeds 1, yet the output grows without bound. This motivates a formal notion of well-behaved systems.

A signal $x[n]$ is **bounded** if there exists a finite constant $B$ such that $|x[n]| \leq B$ for all $n$. A system is **bounded-input bounded-output (BIBO) stable** if every bounded input produces a bounded output. For an LTI system, BIBO stability holds if and only if the impulse response is **absolutely summable**:

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty. \tag{4}$$

To see why this is sufficient, suppose $|x[n]| \leq B$ for all $n$. Then

$$|y[n]| = \left| \sum_{k=-\infty}^{\infty} h[k]\, x[n-k] \right| \leq \sum_{k=-\infty}^{\infty} |h[k]|\, |x[n-k]| \leq B \sum_{k=-\infty}^{\infty} |h[k]|,$$

which is finite, so the output is bounded.

The accumulator with impulse response $h[n] = u[n]$ fails this condition, as the infinite summation of its impulse response diverges:

$$\sum_{n=-\infty}^{\infty} |u[n]| = \sum_{n=0}^{\infty} 1 \to \infty$$

## 12.4 Frequency Response

The impulse response completely characterizes LTI systems in the time domain via convolution. However, we can use the eigenfunction property of LTI systems to view this same relationship in the frequency domain.

Consider a specific input signal $x[n] = e^{i\omega n}$. Passing this through an LTI system with impulse response $h[n]$ using convolution yields:

$$\begin{aligned} y[n] = (h * x)[n] &= \sum_{k=-\infty}^{\infty} h[k] x[n-k] \\ &= \sum_{k=-\infty}^{\infty} h[k] e^{i\omega(n-k)} \\ &= \left( \sum_{k=-\infty}^{\infty} h[k] e^{-i\omega k} \right) e^{i\omega n}. \end{aligned}$$

The term in the parentheses is independent of time $n$. We define this scaling factor as the **frequency response** $H(\omega)$:

$$H(\omega) = \sum_{k=-\infty}^{\infty} h[k] e^{-i\omega k}. \tag{5}$$

Thus, for an input $e^{i\omega n}$, the output is simply $H(\omega)e^{i\omega n}$. The complex exponential passes through the system unchanged except for multiplication by the complex number $H(\omega)$. Because of this, complex exponentials are called **eigenfunctions** of LTI systems, with $H(\omega)$ as the eigenvalue for frequency $\omega$.

**When does $H(\omega)$ exist?** The sum in (5) converges for every $\omega$ whenever $\sum_{k=-\infty}^{\infty} |h[k]| < \infty$, which is precisely the BIBO stability condition (4). For the accumulator $h[n] = u[n]$ the sum $\sum_{k=0}^{\infty} e^{-i\omega k}$ does not converge, so the frequency response is not defined by this formula. For the rest of this section we assume BIBO stable systems.

### 12.4.1 Frequency response of general signals

The same idea extends to general signals by superposition. Suppose a signal is a weighted combination of complex exponentials, $x[n] = \sum_m c_m e^{i\omega_m n}$. Each term is scaled independently by the system:

$$x[n] = \sum_m c_m e^{i\omega_m n} \implies y[n] = \sum_m c_m H(\omega_m) e^{i\omega_m n}.$$

Defining $X(\omega)$ and $Y(\omega)$ as the frequency-domain representations of the input and output respectively, this relationship can be written compactly as

$$Y(\omega) = H(\omega)X(\omega). \tag{6}$$

That is, the output at each frequency is simply the input at that frequency scaled by $H(\omega)$. Since we also have that $y = x * h$ in the time domain, we conclude that **convolution in time corresponds to multiplication in frequency**. An LTI system is therefore determined either by $h[n]$ in time or by $H(\omega)$ in frequency.

### 12.4.2 Interpretation

- **Magnitude** $|H(\omega)|$ shows which frequencies are passed, attenuated, or boosted. For a smoothing filter, $|H(\omega)|$ has lower values for higher frequencies.
- **Phase** $\angle(H(\omega))$ sets timing and waveform shape. Linear phase corresponds to pure delay and avoids waveform distortion.

### 12.4.3 Extensions

While the frequency response we derived above provides a theoretical framework for aperiodic, infinite-length sequences, practical computation requires finite-length data. For this, we use the $N$-point DFT, which inherently assumes the time-domain sequence is periodic with period $N$. Consequently, element-wise multiplication in the DFT domain corresponds to circular convolution in time.

To recover standard convolution for signals of lengths $L$ and $M$, we must zero-pad inputs to a length $N \geq L + M - 1$ prior to the DFT. This padding prevents time-domain aliasing (wrap-around). Efficiency is achieved via the Fast Fourier Transform (FFT), which reduces complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(N \log N)$. For long streaming signals, block processing algorithms like overlap-add or overlap-save utilize this FFT-based convolution to process data in chunks.